

Experiment No. 4

Title: Use Function Types, Scope and Closures; Apply Try-Catch; Build a Palindrome Checker

Software / Tools Required:

1. Visual Studio Code
2. Google Chrome
3. HTML5
4. JavaScript (ES6) Theory:

JavaScript provides different types of functions and scope mechanisms that make programs modular and reusable. Exception handling helps in managing runtime errors effectively. a)

Function Types

1. Functions perform specific tasks.
2. JavaScript supports function declarations, function expressions, and arrow functions. b)

Scope

1. Scope determines the accessibility of variables.
2. Types include global scope, function scope, and block scope.

c) Closures

1. A closure is a function that remembers variables from its outer scope.
2. It allows access to outer variables even after the outer function has finished execution. d)

Try-Catch

1. Used for exception handling.
2. Prevents program termination by catching runtime errors.

e) Palindrome Checker

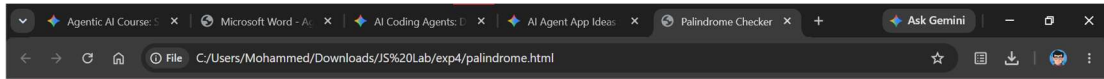
1. A palindrome is a word or number that reads the same forward and backward.
2. JavaScript can compare the original and reversed string to determine whether it is a palindrome.

Case Study Implemented:

A vehicle registration number validation webpage was developed using JavaScript to check whether the entered registration number is valid or not. A palindrome checker was also implemented to verify whether the entered text is a palindrome.

Conclusion:

The practical was completed successfully. Function types, scope, closures, and try-catch exception handling were implemented successfully. A vehicle registration number validator and a palindrome checker were developed using JavaScript to validate user input and demonstrate string manipulation concepts.

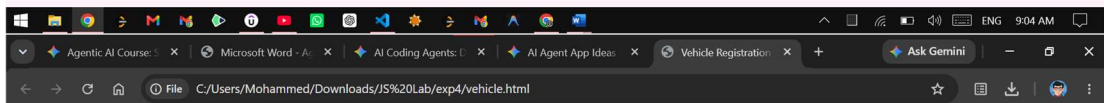


Palindrome Checker

Enter a word, phrase, or number to check if it's a palindrome.

Check

"abbba" is a Palindrome!



Vehicle Validator

Enter an Indian vehicle registration number (e.g., MH 01 AB 1234)

Validate

"MH25CU1234" is a VALID registration number!